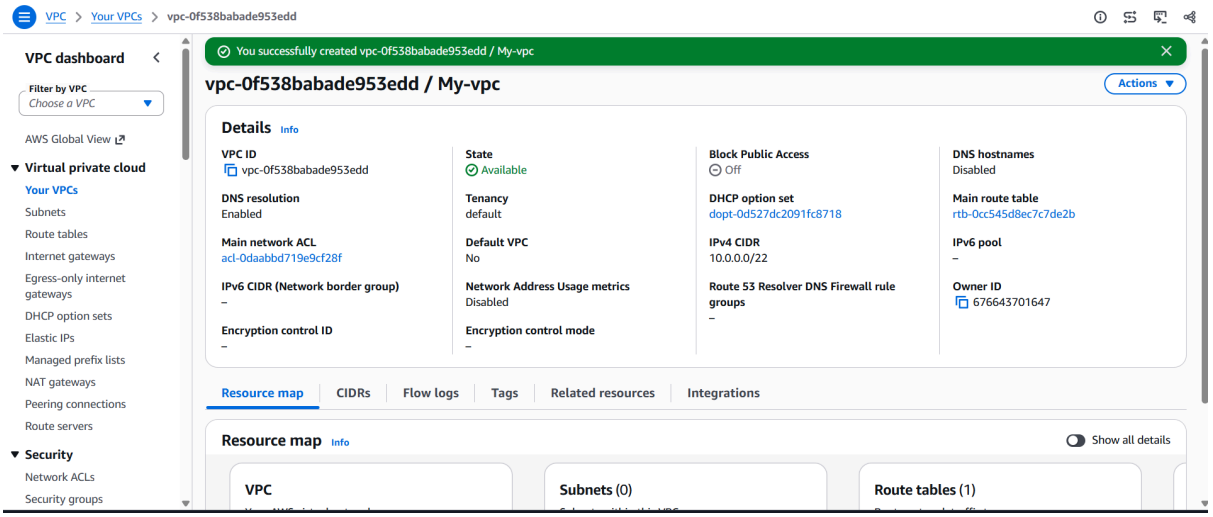
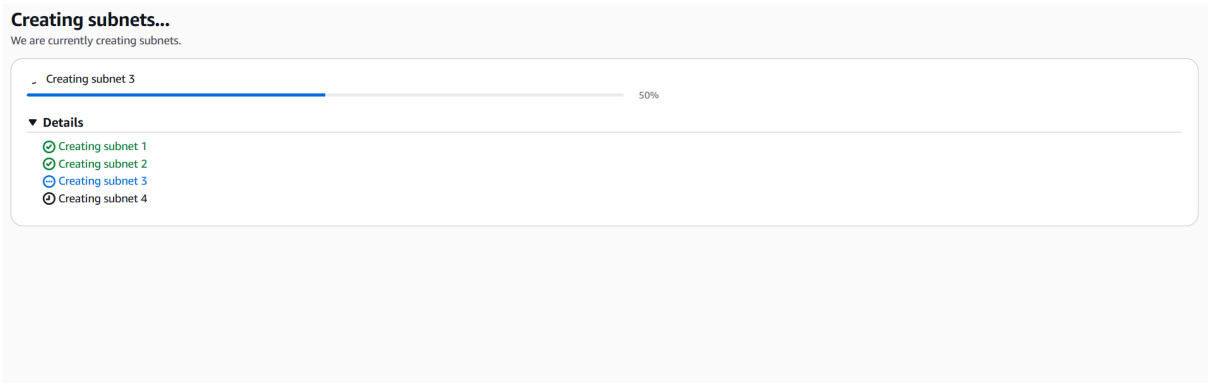
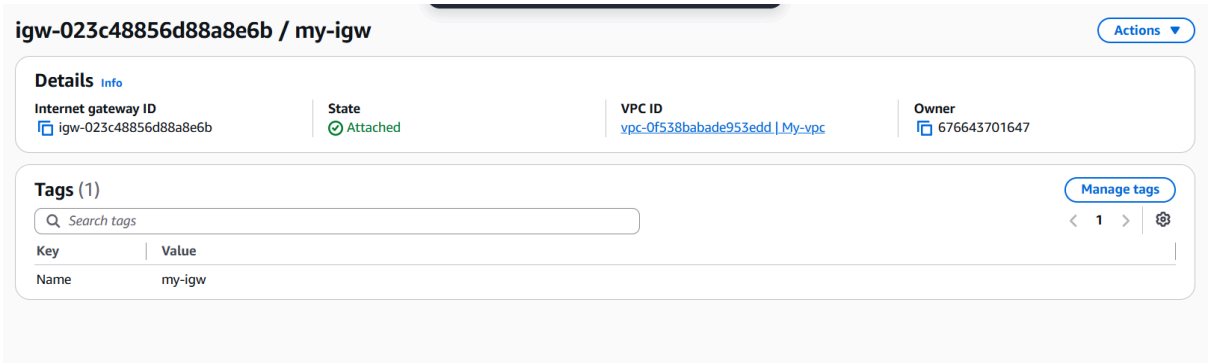
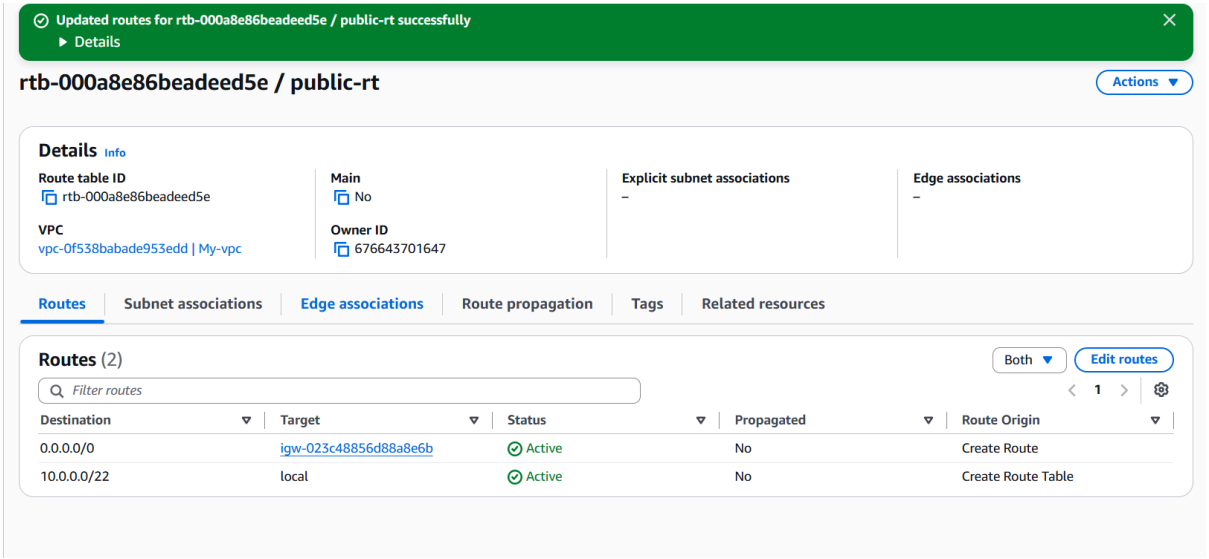
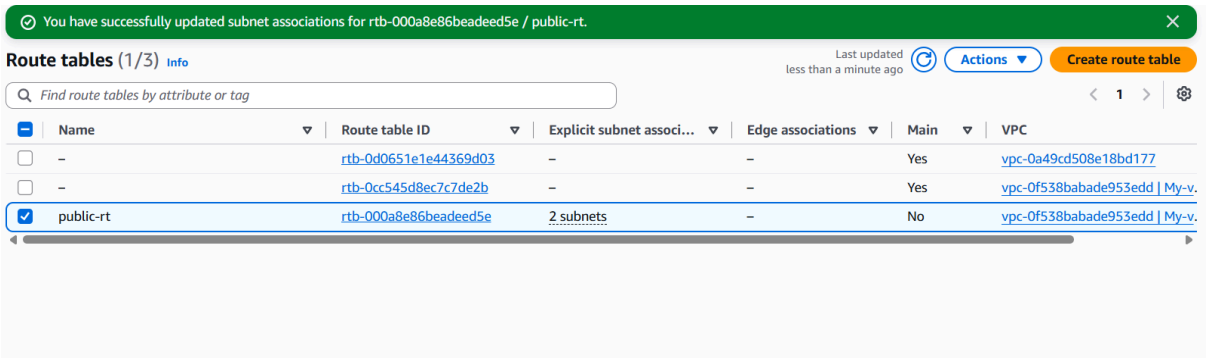
1. Create VPC with 2 private and 2 public subnets.

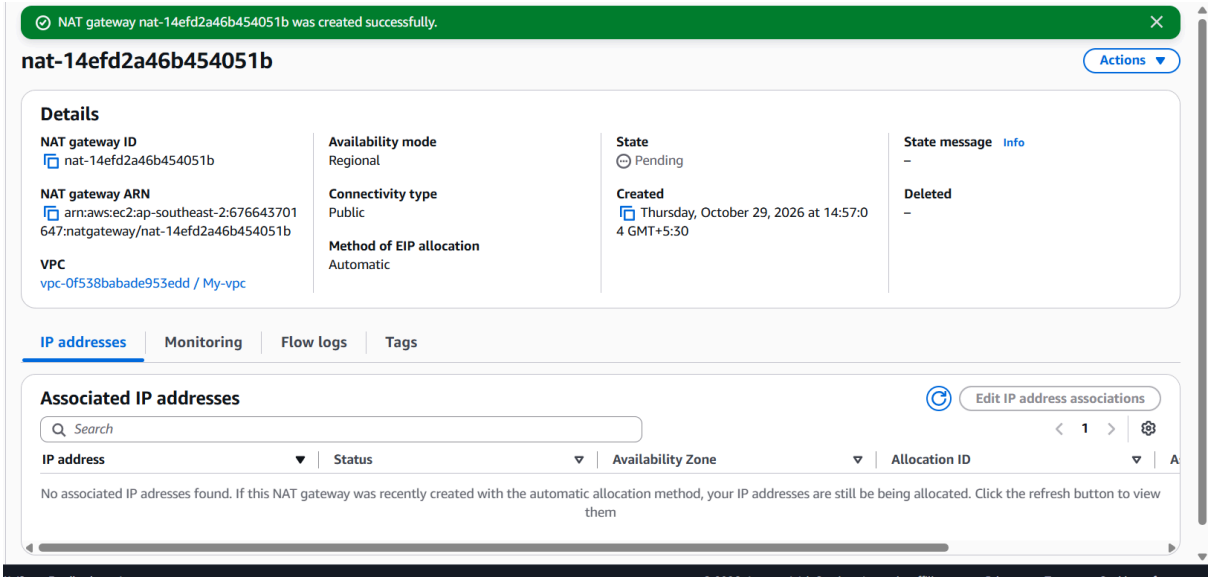


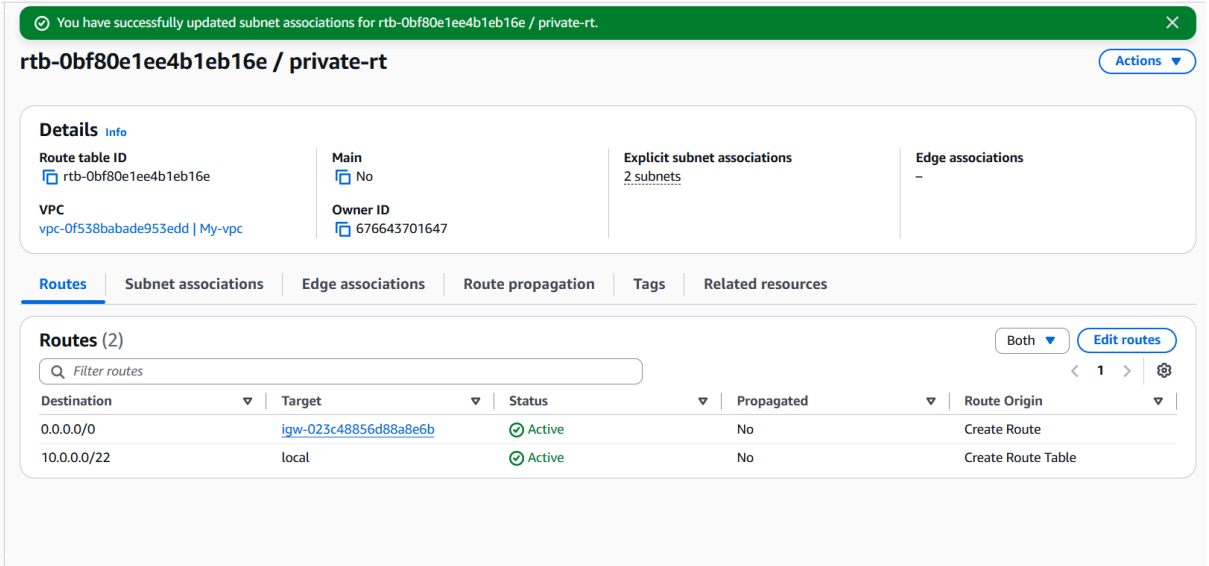





Updated routes for rtb-000a8e86beadeed5e / public-rt successfully
Details
rtb-000a8e86beadeed5e / public-rt
Actions
Details Info
Route table ID
rtb-000a8e86beadeed5e
Main
No
Explicit subnet associations
–
Edge associations
–
VPC
vpc-0f538babade953edd | My-vpc
Owner ID
676643701647
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Routes (2)
Both
Edit routes
Filter routes
1
Destination
Target
Status
Propagated
Route Origin
0.0.0.0/0
igw-023c48856d88a8e6b
Active
No
Create Route
10.0.0.0/22
local
Active
No
Create Route Table

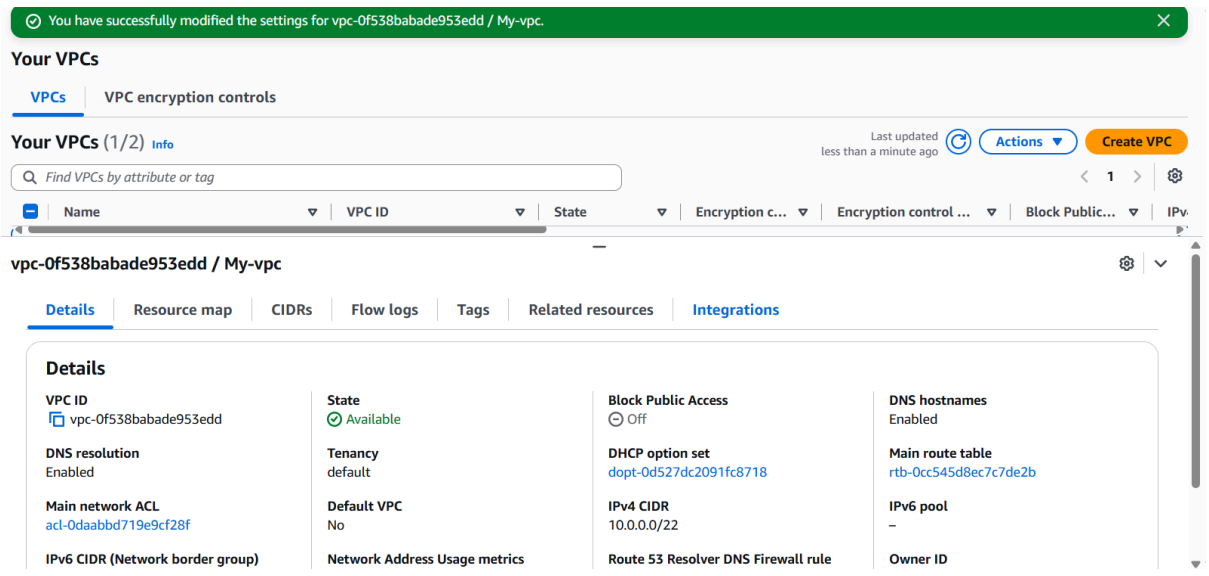

You have successfully updated subnet associations for rtb-000a8e86beadeed5e / public-rt.
Route tables (1/3) Info
Last updated
less than a minute ago
Actions
Create route table
Find route tables by attribute or tag
1
Name
Route table ID
Explicit subnet associ...
Edge associations
Main
VPC
–
rtb-0d0651e1e44369d03
–
–
Yes
vpc-0a49cd508e18bd177
–
rtb-0cc545d8ec7c7de2b
–
–
Yes
vpc-0f538babade953edd | My-v.
public-rt
rtb-000a8e86beadeed5e
2 subnets
–
No
vpc-0f538babade953edd | My-v.


NAT gateway nat-14efd2a46b454051b was created successfully.
nat-14efd2a46b454051b
Actions
Details
NAT gateway ID
nat-14efd2a46b454051b
NAT gateway ARN
arn:aws:ec2:ap-southeast-2:676643701647:natgateway/nat-14efd2a46b454051b
VPC
vpc-0f538babade953edd / My-vpc
Availability mode
Regional
Connectivity type
Public
Method of EIP allocation
Automatic
State
Pending
Created
Thursday, October 29, 2026 at 14:57:04 GMT+5:30
State message Info
–
Deleted
–
IP addresses
Monitoring
Flow logs
Tags
Associated IP addresses
Edit IP address associations
Search
1
IP address
Status
Availability Zone
Allocation ID
No associated IP adresses found. If this NAT gateway was recently created with the automatic allocation method, your IP addresses are still be being allocated. Click the refresh button to view them



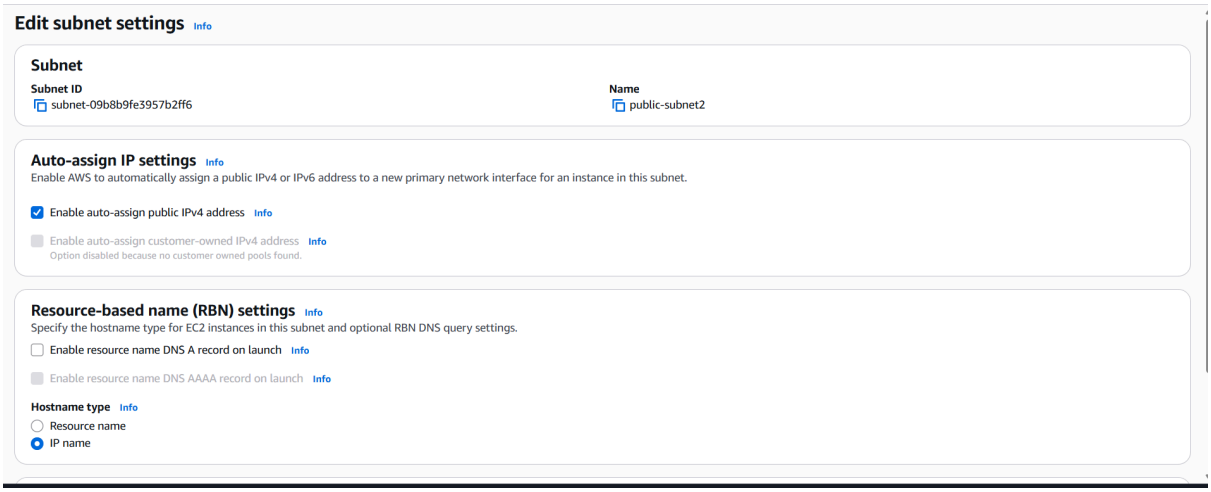
1. I created a VPC named **My-vpc** with CIDR block `10.0.0.0/22`.
2. I created two public subnets
3. I created two private subnets
4. I created an **Internet Gateway** and attached it to my VPC.
5. I created a **public route table** and added a route to the Internet Gateway.
6. I associated both public subnets with the public route table.
7. I created a **NAT Gateway** in one of my public subnets and allocated an Elastic IP.
8. I waited until the NAT Gateway became **Available**.
9. I created a **private route table** and added a route through the NAT Gateway.
10. I associated both private subnets with the private route table.
11. I verified that my VPC contains **2 public subnets and 2 private subnets** with the required routing configuration.

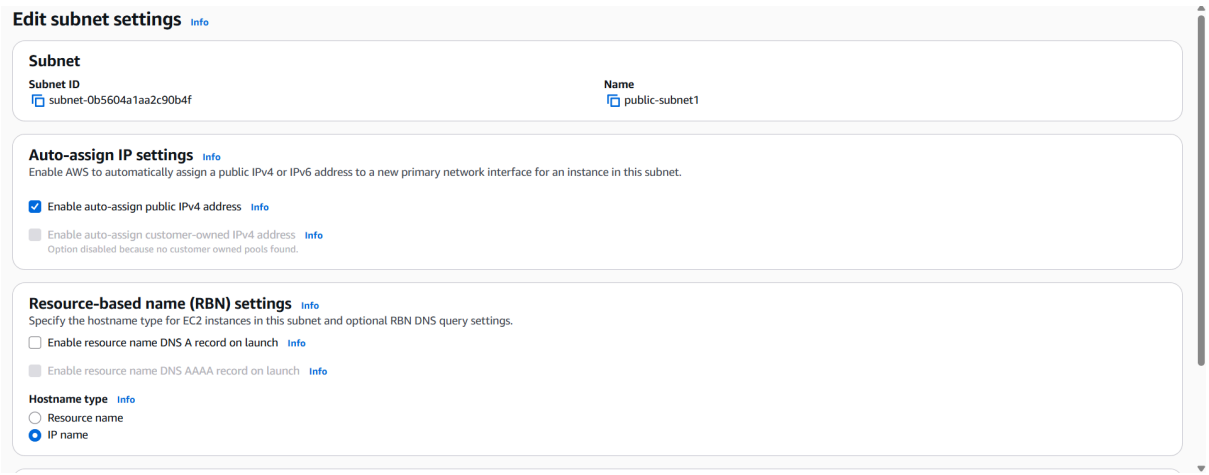
2.Enable DNS Hostname in VPC



1. I opened the AWS VPC Console.
2. I selected Your VPCs.
3. I selected my My-vpc.
4. I clicked Actions → Edit VPC settings.
5. I enabled DNS hostnames.
6. I clicked Save changes.
7. I verified that DNS hostnames is enabled for my VPC.

3. Enable Auto Assign Public IP in 2 public subnets

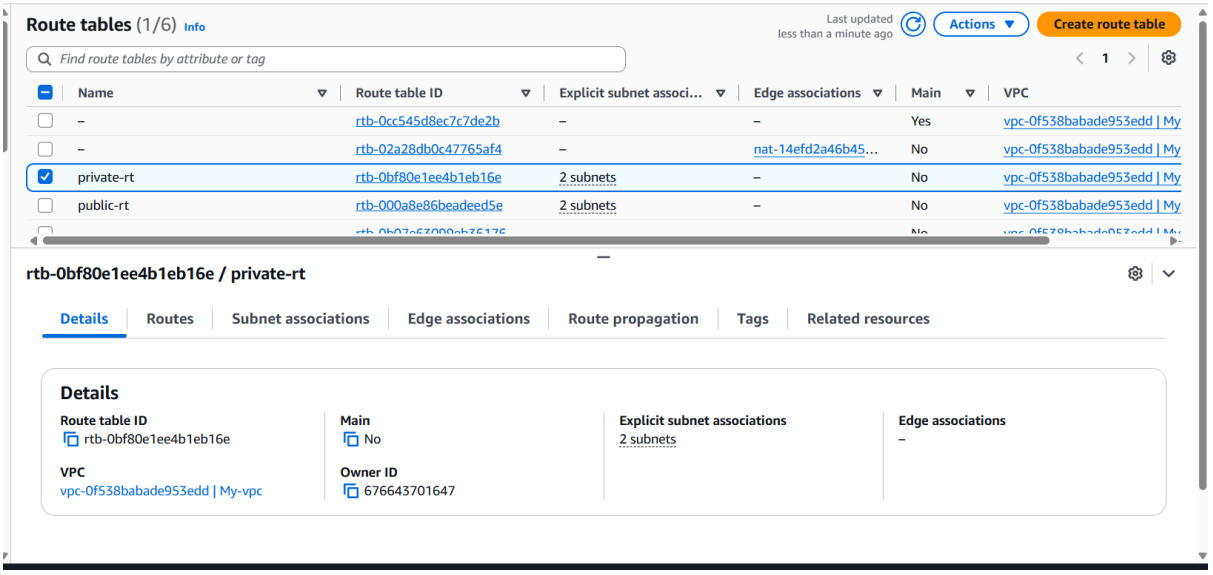




1. I opened VPC → Subnets.
2. I selected public-subnet1.
3. I clicked Actions → Edit subnet settings.
4. I enabled Auto-assign public IPv4 address.
5. I clicked Save.

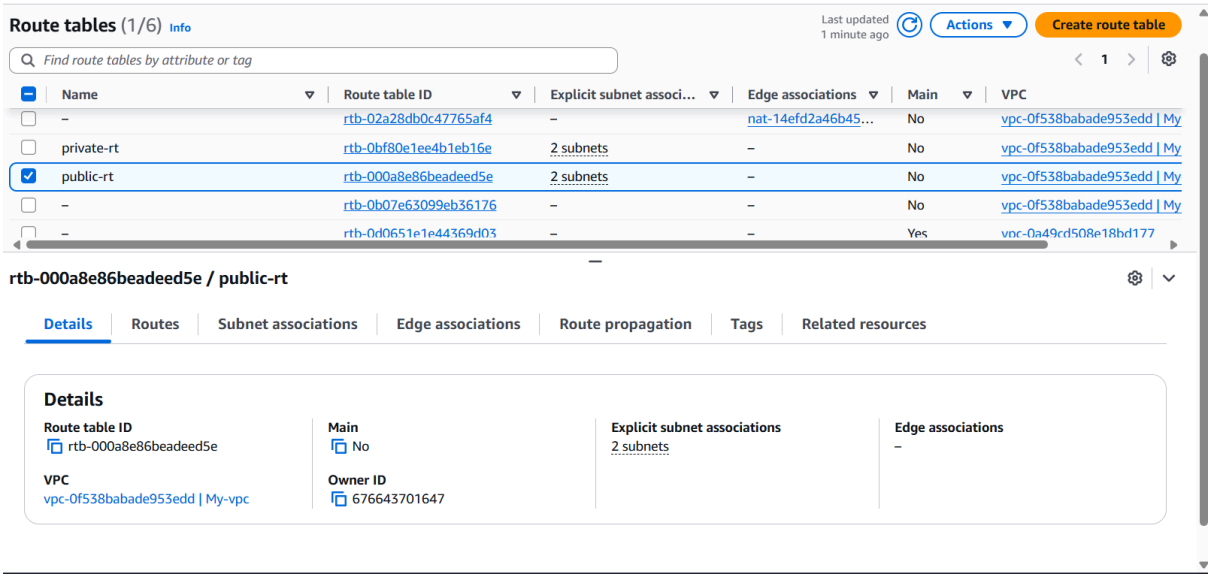
6. I selected public-subnet2.
7. I clicked Actions → Edit subnet settings.
8. I enabled Auto-assign public IPv4 address.
9. I clicked Save.
10. I verified that public IP auto-assignment is enabled for both public subnets.

4. Add 2 private subnets in private route table



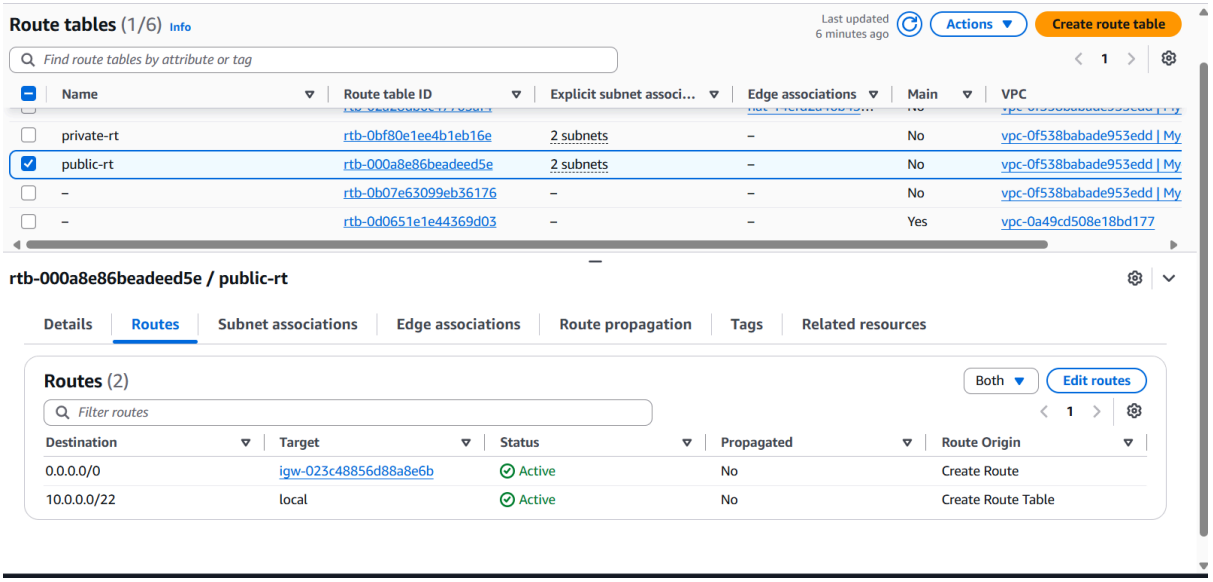
1. I opened **VPC → Route Tables**.
2. I selected my **private route table (`private-rt`)**.
3. I opened the **Subnet associations** tab.
4. I clicked **Edit subnet associations**.
5. I selected **private-subnet1**.
6. I selected **private-subnet2**.
7. I clicked **Save associations**.
8. I verified that both private subnets are associated with the private route table.

5. Add 2 public subnets in public route table



1. I opened VPC → Route Tables.
2. I selected my public route table.
3. I opened the Subnet associations tab.
4. I clicked Edit subnet associations.
5. I selected public-subnet1.
6. I selected public-subnet2.
7. I clicked Save associations.
8. I verified that both public subnets are associated with the public route table.

6. Public route table will have the routes to internet and local.



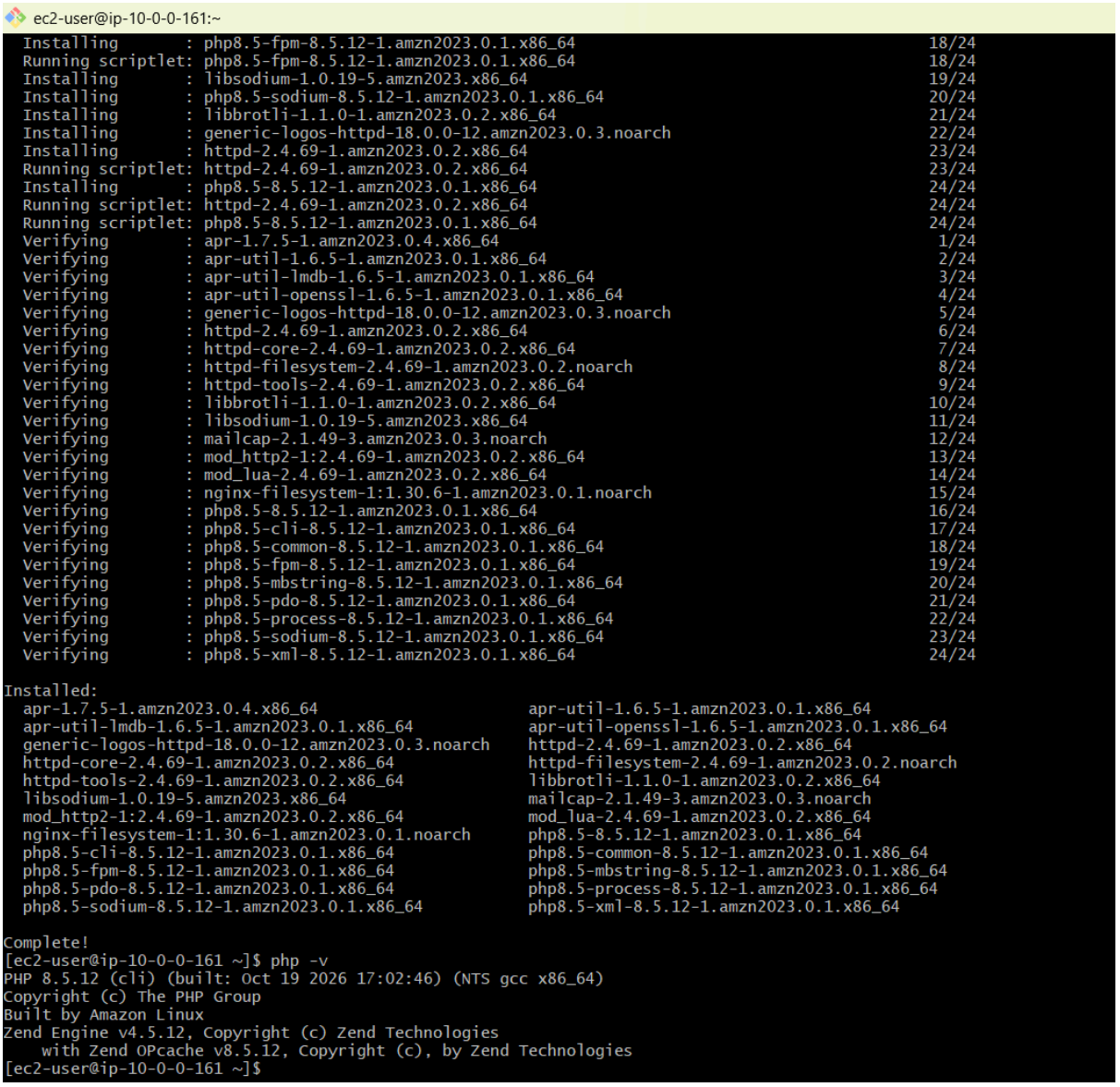
1. I opened **VPC → Route Tables**.
2. I selected my **public route table**.
3. I opened the **Routes** tab.

4. I clicked **Edit routes**.
5. I verified that the **local route** is already present.
6. I clicked **Add route**.
7. For **Destination**, I entered `0.0.0.0/0`.
8. For **Target**, I selected **Internet Gateway**.
9. I selected my Internet Gateway.
10. I clicked **Save changes**.
11. I verified that the public route table contains both **local** and **Internet Gateway** routes.

7. Create EC2 in public subnet with t2.micro and install PHP.

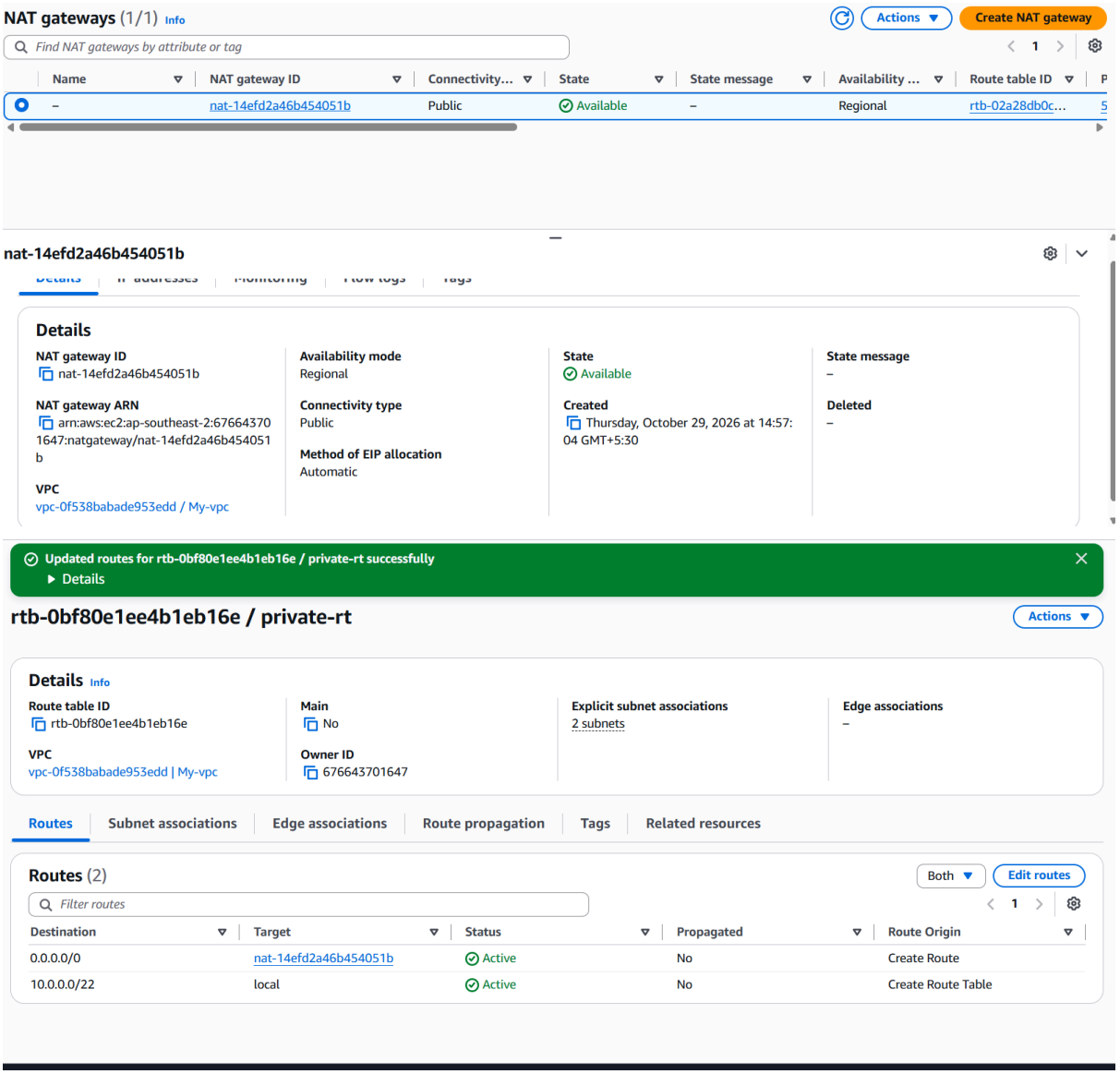
ec2-user@ip-10-0-0-161:~

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/Downloads (master)
$fssh -i linux-testing.pem ec2-user@15.135.232.7
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-0-161 ~]$ cat /etc/os-release
NAME="Amazon Linux"
VERSION="2023"
ID="amzn"
ID_LIKE="fedora"
VERSION_ID="2023"
PLATFORM_ID="platform:al2023"
PRETTY_NAME="Amazon Linux 2023.12.20261028"
ANSI_COLOR="0;33"
CPE_NAME="cpe:2.3:o:amazon:amazon_linux:2023"
HOME_URL="https://aws.amazon.com/linux/amazon-linux-2023/"
DOCUMENTATION_URL="https://docs.aws.amazon.com/linux/"
SUPPORT_URL="https://aws.amazon.com/premiumsupport/"
BUG_REPORT_URL="https://github.com/amazonlinux/amazon-linux-2023"
VENDOR_NAME="AWS"
VENDOR_URL="https://aws.amazon.com/"
SUPPORT_END="2029-06-30"
[ec2-user@ip-10-0-0-161 ~]$ sudo dnf update -y
Amazon Linux 2023 Kernel Livepatch repository                   806 kB/s |  89 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-10-0-0-161 ~]$ sudo dnf install php -y
Last metadata expiration check: 0:00:14 ago on Thu Oct 29 10:23:59 2026.
Dependencies resolved.
================================================================================================
 Package                  Architecture   Version                        Repository          Size
================================================================================================
Installing:
 php8.5                   x86_64         8.5.12-1.amzn2023.0.1          amazonlinux         13 k
Installing dependencies:
 apr                      x86_64         1.7.5-1.amzn2023.0.4           amazonlinux        129 k
 apr-util                 x86_64         1.6.5-1.amzn2023.0.1           amazonlinux        100 k
 apr-util-lmdb            x86_64         1.6.5-1.amzn2023.0.1           amazonlinux         15 k
 generic-logos-httpd      noarch         18.0.0-12.amzn2023.0.3         amazonlinux         19 k
 httpd-core               x86_64         2.4.69-1.amzn2023.0.2          amazonlinux        1.4 M
 httpd-filesystem         noarch         2.4.69-1.amzn2023.0.2          amazonlinux         12 k
```



1. I opened **AWS Console → EC2**.
2. I clicked **Launch instance**.
3. I entered the instance name as **PHP-Server**.
4. I selected an **Amazon Linux** AMI.
5. I selected **t2.micro** as the instance type.
6. I selected my existing **key pair**.
7. I selected my **My-vpc**.
8. I selected **public-subnet1**.
9. I enabled **Auto-assign Public IP**.
10. I configured the security group to allow **SSH (22)** and **HTTP (80)**.
11. I launched the EC2 instance.
12. I waited until the instance was **Running** and passed the status checks.
13. I connected to the EC2 instance using SSH.
14. I installed **PHP** on the EC2 instance.
15. I verified that PHP was installed successfully.

8. Configure NAT gateway in public subnet and connect to private instance



1. I verified that my **NAT Gateway** is in `public-subnet1` and its status is **Available**.
2. I verified that my **private route table** has `0.0.0.0/0` pointing to the NAT Gateway.
3. I verified that `private-subnet1` and `private-subnet2` are associated with the private route table.
4. I opened **EC2 → Launch Instance**.
5. I created a new EC2 instance named **Private-Server**.
6. I selected **Amazon Linux** and **t2.micro**.
7. I selected my existing **key pair**.
8. In **Network settings**, I selected **My-vpc**.
9. I selected **private-subnet1**.

10. I disabled **Auto-assign Public IP**.
11. I configured the security group to allow **SSH (22)** from my public EC2/security group.
12. I launched the private EC2.
13. I waited until the instance was **Running**.
14. I verified that the private EC2 has a **Private IPv4 address** and no public IPv4 address.
15. I connected to my **public EC2**.
16. From the public EC2, I connected to the private EC2 using its **private IP address**.
17. I verified internet connectivity from the private EC2 through the **NAT Gateway**.

9. Install Apache Tomcat in private EC2 and deploy a sample app

ec2-user@ip-10-0-0-161:/tmp

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~ (master)
$ cd Downloads

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/Downloads (master)
$ ssh -i linux-testing.pem ec2-user@10.0.0.161
ssh: connect to host 10.0.0.161 port 22: Connection timed out

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/Downloads (master)
$ ssh -i linux-testing.pem ec2-user@15.135.232.7
       #_
   ,    ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Thu Oct 29 10:45:34 2026 from 103.203.172.230
[ec2-user@ip-10-0-0-161 ~]$ ssh ec2-user@10.0.1.134
The authenticity of host '10.0.1.134 (10.0.1.134)' can't be established.
ED25519 key fingerprint is SHA256:TuxJPNyPyBGYy4jkFh55dDknEYsjOdHSuwNVzISHnAc.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.1.134' (ED25519) to the list of known hosts.
ec2-user@10.0.1.134: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[ec2-user@ip-10-0-0-161 ~]$ curl -I https://www.google.com
HTTP/2 200
content-type: text/html; charset=ISO-8859-1
content-security-policy-report-only: object-src 'none';base-uri 'self';script-src 'nonce-lWgkTuAvffUnyl
ps3ibcJg' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http:;report-uri https:
//csp.withgoogle.com/csp/gws/other-hp
accept-ch: Sec-CH-Prefers-Color-Scheme
p3p: CP="This is not a P3P policy! See g.co/p3phelp for more info."
date: Thu, 29 Oct 2026 11:02:01 GMT
server: gws
x-xss-protection: 0
x-frame-options: SAMEORIGIN
expires: Thu, 29 Oct 2026 11:02:01 GMT
cache-control: private
set-cookie: __Secure-STRP=ABNhA6jcP3pJb8obOgsVHRtAZDLDrQdMPCt_G3JZClHLqKzYN29o-9qUmqtLzIrlXKh8t1bFb11Ok
eGuRxAJVSzrbR3NempTOWfP; expires=Thu, 29-Oct-2026 11:07:01 GMT; path=/; domain=.google.com; Secure; Sam
eSite=strict
set-cookie: AEC=Aaa9EJqNApoYkBzOzW0aY4DFIWoiwoZz0tB1BtnXnYStgkdR4FxM2ehYdQ; expires=Tue, 27-Apr-2027 11
:02:01 GMT; path=/; domain=.google.com; Secure; HttpOnly; SameSite=lax
set-cookie: NID=Cu4BCAESoAEBM9AglzrPn6fDTfGvJzU3t1ju1x1gUjw9RNzXio7XjLDda11XMumsluQ6frGsctPHlHCjqrWsAO7
rzcKooTA8aQ_TY_pc1LaZAFfuAjsVVWJZwS-gRjZmCNy-3lUo-BNz2rO3weanYMYZHjScBvmOR7HO-HZZOBrVR_KadxVlBIIgjcZnQF
COAaVrqDpFvHByRR9bGaSdKmUaxWs1dtFhKAEyRQFwSkLPoXaJq-EjbAZ1haQRAIEOCBrxdBC3cHJB9XfvPxjpcFPd5yANhS4ySfU7R
KBHp-MfKCW8yqO8OYPZqr-v6lSRCg; expires=Fri, 30-Apr-2027 11:02:01 GMT; path=/; domain=.google.com; HttpO
nly
alt-svc: h3=":443"; ma=2592000,h3-29=":443"; ma=2592000

[ec2-user@ip-10-0-0-161 ~]$ sudo dnf install java-17-amazon-corretto -y
Last metadata expiration check: 0:38:12 ago on Thu Oct 29 10:23:59 2026.
Dependencies resolved.
================================================================================================
```

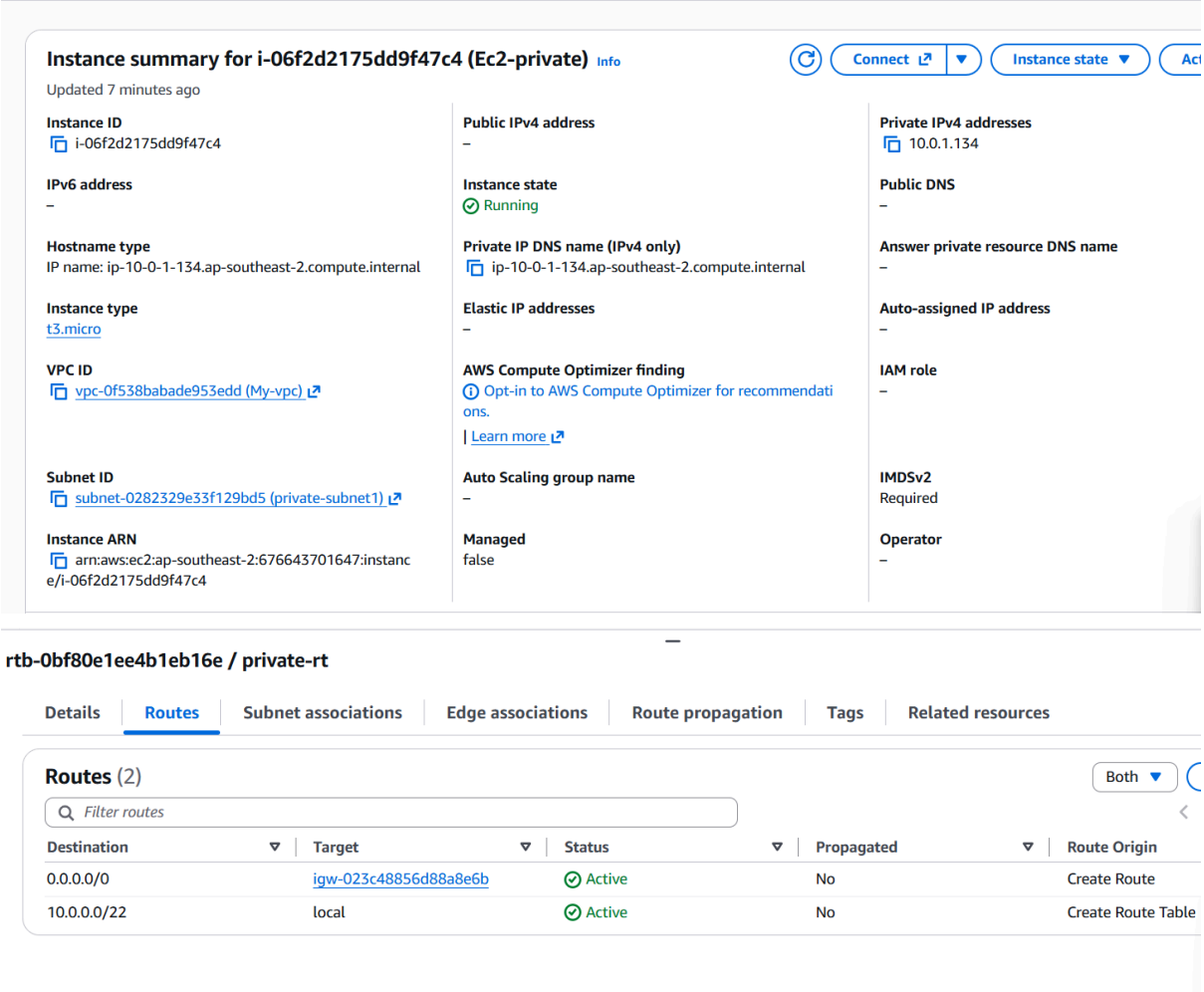
ec2-user@ip-10-0-0-161:/tmp

```
            </div>
            <div id="middle" class="curved container">
                <h3>Developer Quick Start</h3>
                <div class="col25">
                    <div class="container">
                        <p><a href="/docs/setup.html">Tomcat Setup</a></p>
                        <p><a href="/docs/appdev/">First Web Application</a></p>
                    </div>
                </div>
                <div class="col25">
                    <div class="container">
                        <p><a href="/docs/realm-howto.html">Realms &amp;amp; AAA</a></p>
                        <p><a href="/docs/jndi-datasource-examples-howto.html">JDBC DataSources</a></p>
                    </div>
                </div>
                <div class="col25">
                    <div class="container">
                        <p><a href="/examples/">Examples</a></p>
                    </div>
                </div>
                <div class="col25">
                    <div class="container">
                        <p><a href="https://cwiki.apache.org/confluence/display/TOMCAT/Specifications">
Servlet Specifications</a></p>
                        <p><a href="https://cwiki.apache.org/confluence/display/TOMCAT/Tomcat+Versions"
>Tomcat Versions</a></p>
                    </div>
                </div>
                <br class="separator" />
            </div>
            <div id="lower">
                <div id="low-manage" class="">
                    <div class="curved container">
                        <h3>Managing Tomcat</h3>
                        <p>For security, access to the <a href="/manager/html">manager webapp</a> is re
stricted.
                        Users are defined in:</p>
                        <pre>$CATALINA_HOME/conf/tomcat-users.xml</pre>
                        <p>In Tomcat 10.1 access to the manager application is split between
                           different users. &amp;nbsp; <a href="/docs/manager-howto.html">Read more...</a><
/p>
                        <br />
                        <h4><a href="/docs/RELEASE-NOTES.txt">Release Notes</a></h4>
                        <h4><a href="/docs/changelog.html">Changelog</a></h4>
                        <h4><a href="https://tomcat.apache.org/migration.html">Migration Guide</a></h4>
                        <h4><a href="https://tomcat.apache.org/security.html">Security Notices</a></h4>
                    </div>
                </div>
                <div id="low-docs" class="">
                    <div class="curved container">
                        <h3>Documentation</h3>
                        <h4><a href="/docs/">Tomcat 10.1 Documentation</a></h4>
                        <h4><a href="/docs/config/">Tomcat 10.1 Configuration</a></h4>
                        <h4><a href="https://cwiki.apache.org/confluence/display/TOMCAT/">Tomcat Wiki</
a></h4>
                        <p>Find additional important configuration information in:</p>
                        <pre>$CATALINA_HOME/RUNNING.txt</pre>
                        <p>Developers may be interested in:</p>
```

ec2-user@ip-10-0-0-161:/tmp

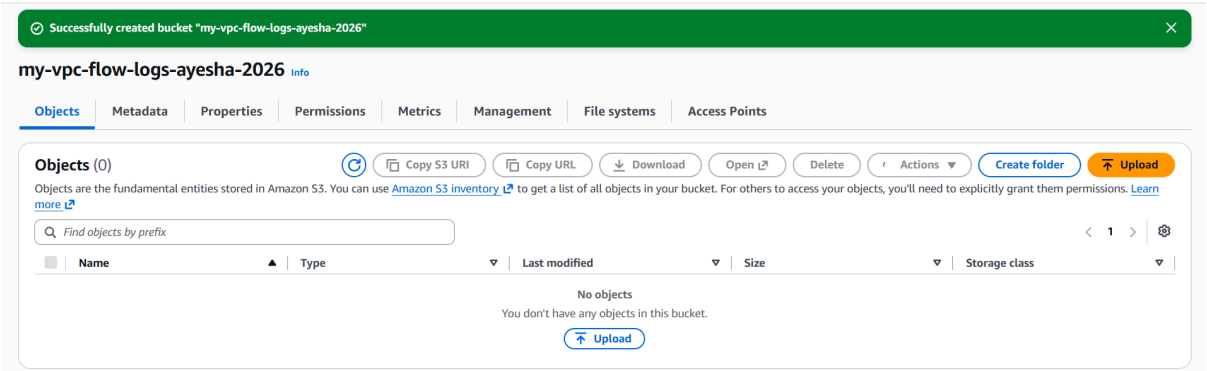
```
                <div class="col20">
                    <div class="container">
                        <h4>Get Involved</h4>
                        <ul>
                            <li><a href="https://tomcat.apache.org/getinvolved.html">Overview</a></li>
                            <li><a href="https://tomcat.apache.org/source.html">Source Repositories</a>
</li>
                            <li><a href="https://tomcat.apache.org/lists.html">Mailing Lists</a></li>
                            <li><a href="https://cwiki.apache.org/confluence/display/TOMCAT/">Wiki</a><
/li>
                        </ul>
                    </div>
                </div>
                <div class="col20">
                    <div class="container">
                        <h4>Miscellaneous</h4>
                        <ul>
                            <li><a href="https://tomcat.apache.org/contact.html">Contact</a></li>
                            <li><a href="https://tomcat.apache.org/legal.html">Legal</a></li>
                            <li><a href="https://www.apache.org/foundation/sponsorship.html">Sponsorshi
p</a></li>
                            <li><a href="https://www.apache.org/foundation/thanks.html">Thanks</a></li>
                        </ul>
                    </div>
                </div>
                <div class="col20">
                    <div class="container">
                        <h4>Apache Software Foundation</h4>
                        <ul>
                            <li><a href="https://tomcat.apache.org/whoweare.html">Who We Are</a></li>
                            <li><a href="https://tomcat.apache.org/heritage.html">Heritage</a></li>
                            <li><a href="https://www.apache.org">Apache Home</a></li>
                            <li><a href="https://tomcat.apache.org/resources.html">Resources</a></li>
                        </ul>
                    </div>
                </div>
                <br class="separator" />
            </div>
            <p class="copyright">Copyright &copy;1999-2026 Apache Software Foundation.  All Rights Rese
rved</p>
        </div>
    </body>

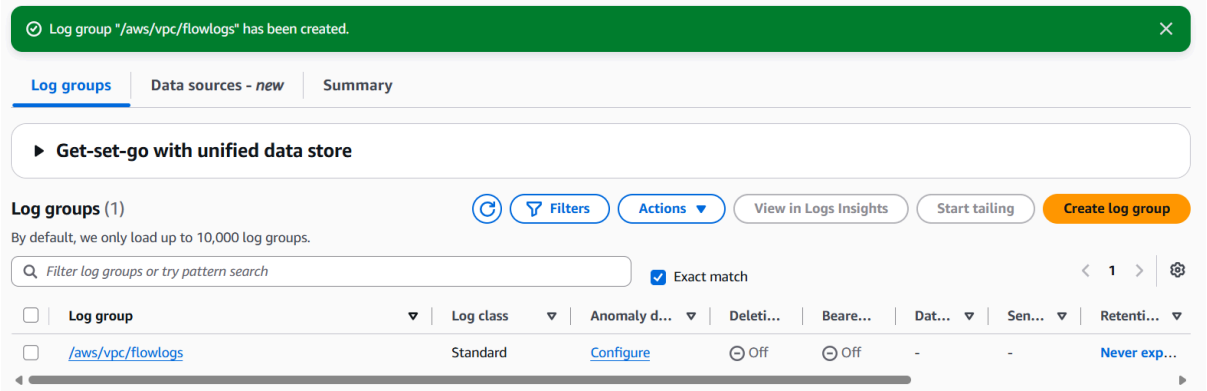
</html>
[ec2-user@ip-10-0-0-161 tmp]$ sudo mkdir -p /opt/tomcat/webapps/sample
[ec2-user@ip-10-0-0-161 tmp]$ sudo vi /opt/tomcat/webapps/sample/index.jsp
[ec2-user@ip-10-0-0-161 tmp]$ sudo chown -R tomcat:tomcat /opt/tomcat/webapps/sample
[ec2-user@ip-10-0-0-161 tmp]$ curl http://localhost:8080/sample/
<html>
<head>
    <title>Sample Application</title>
</head>
<body>
    <h1>Hello from Private EC2</h1>
    <p>Apache Tomcat is running successfully.</p>
</body>
</html>
[ec2-user@ip-10-0-0-161 tmp]$
```

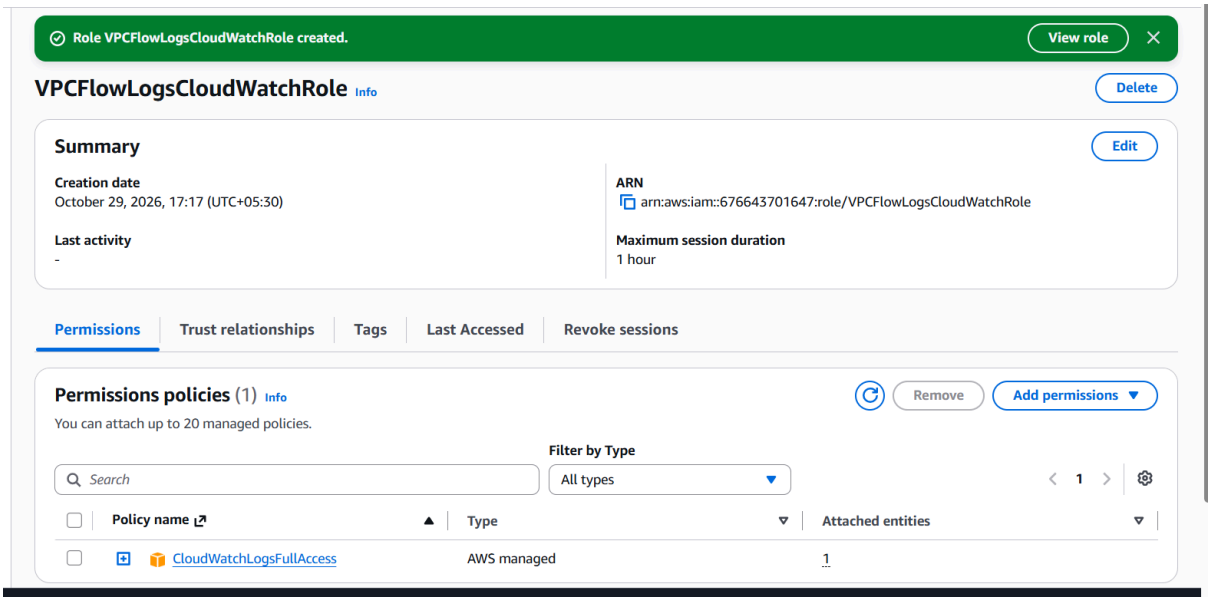


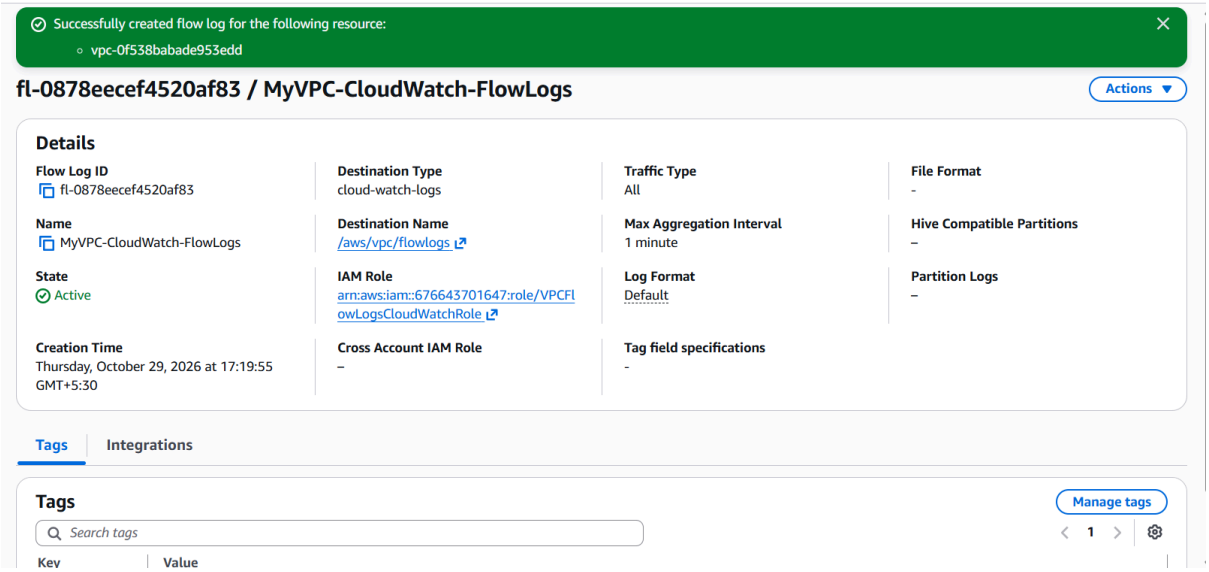
1. I connected to the private EC2 through the public EC2 using SSH.
2. I verified that the private EC2 had internet access through the NAT Gateway.
3. I installed Java 17 on the private EC2.
4. I downloaded and extracted Apache Tomcat into the `/opt` directory.
5. I created a dedicated `tomcat` user and assigned ownership of the Tomcat directory to it.
6. I started the Apache Tomcat server on port `8080`.
7. I verified that Tomcat was running successfully using the local server address.
8. I created a `sample` application directory inside the Tomcat `webapps` directory.
9. I created a sample JSP application and deployed it under the `sample` application directory.
10. I assigned the required ownership to the sample application files.
11. I tested the sample application using the Tomcat port and verified that the application was running successfully.

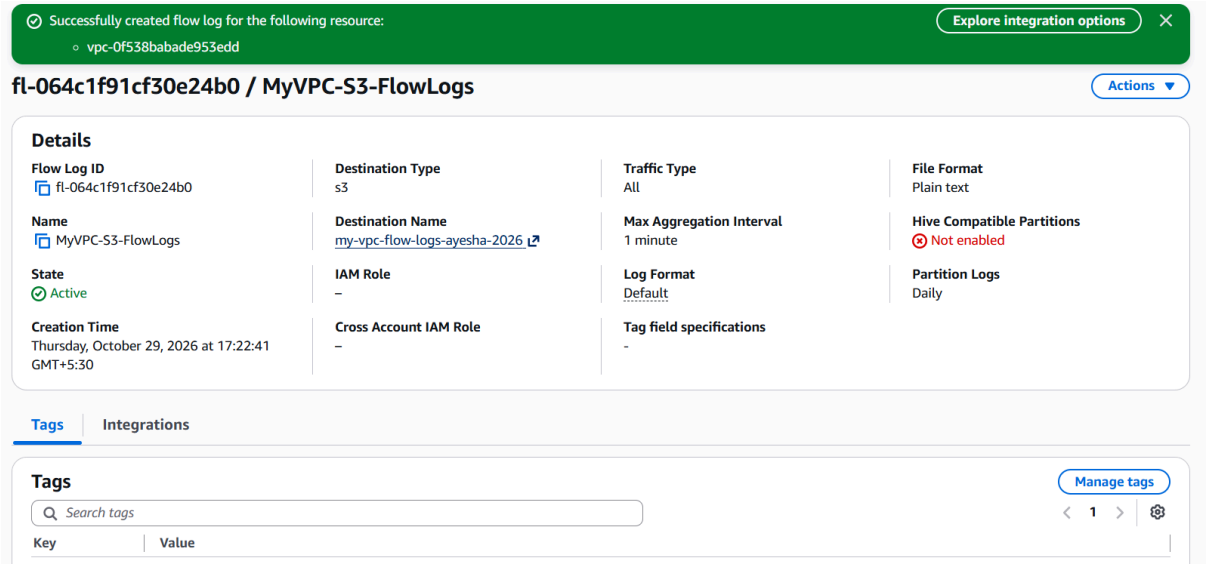
10.Configure VPC flow logs and store the logs in S3 and CloudWatch.

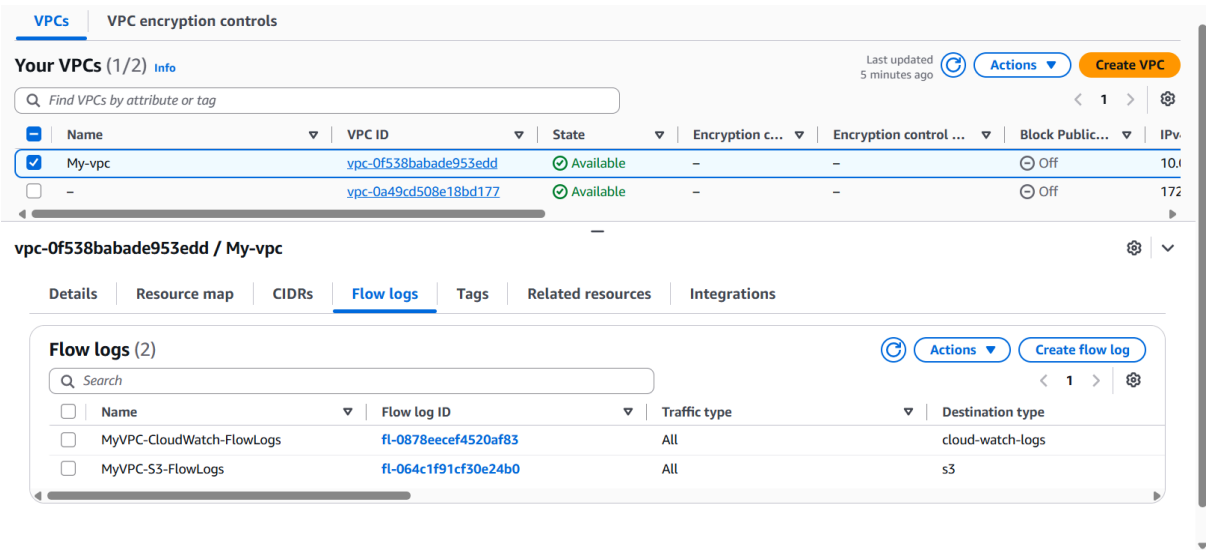

Successfully created bucket "my-vpc-flow-logs-ayesha-2026"
my-vpc-flow-logs-ayesha-2026 Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems
Access Points
Objects (0)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
No objects
You don't have any objects in this bucket.
Upload

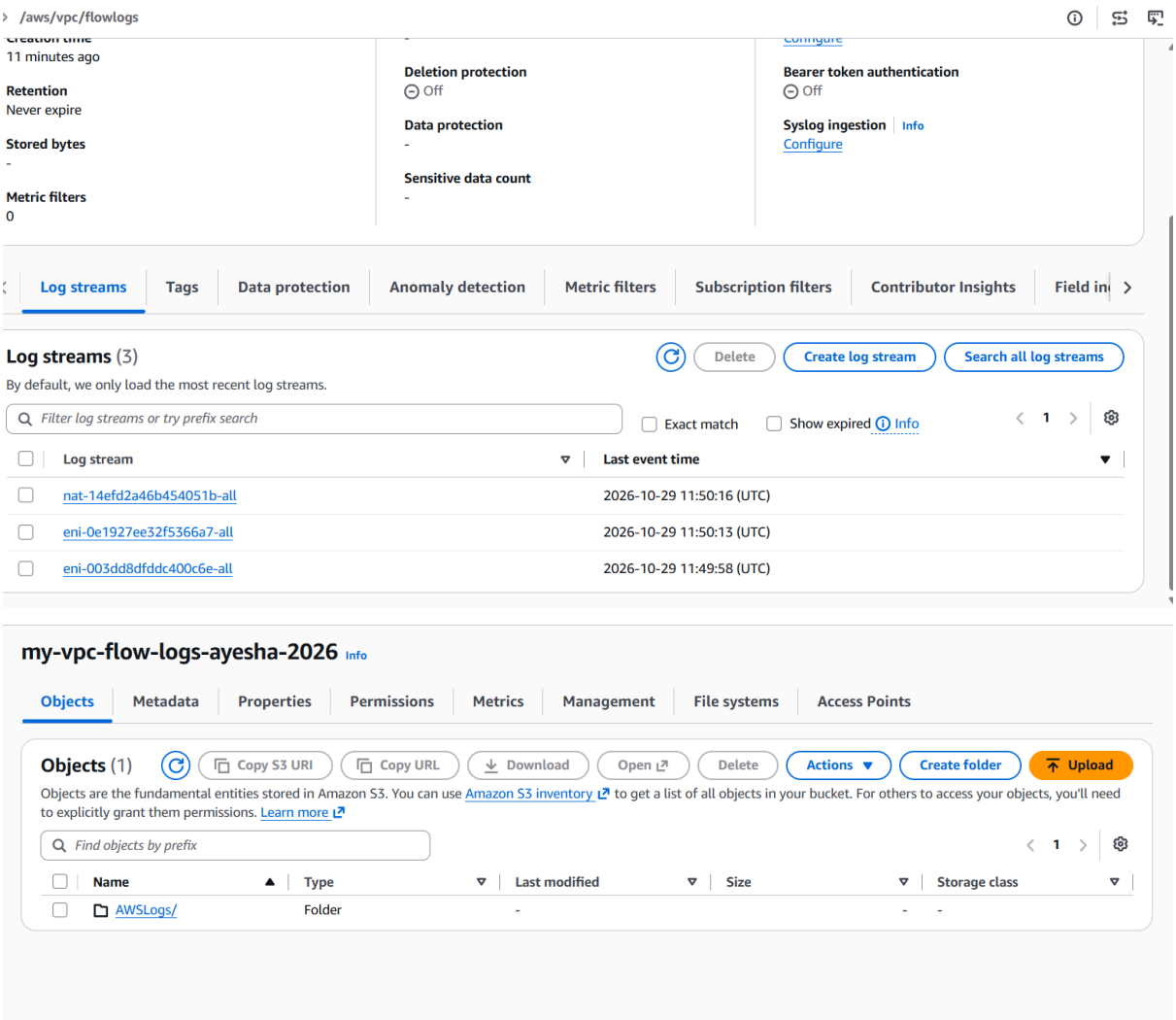

Log group "/aws/vpc/flowlogs" has been created.
Log groups
Data sources - new
Summary
Get-set-go with unified data store
Log groups (1)
Filters
Actions
View in Logs Insights
Start tailing
Create log group
By default, we only load up to 10,000 log groups.
Filter log groups or try pattern search
Exact match
Log group
Log class
Anomaly d...
Deleti...
Beare...
Dat...
Sen...
Retenti...
/aws/vpc/flowlogs
Standard
Configure
Off
Off
-
-
Never exp...


Role VPCFlowLogsCloudWatchRole created.
View role
VPCFlowLogsCloudWatchRole Info
Delete
Summary
Edit
Creation date
October 29, 2026, 17:17 (UTC+05:30)
ARN
arn:aws:iam::676643701647:role/VPCFlowLogsCloudWatchRole
Last activity
-
Maximum session duration
1 hour
Permissions
Trust relationships
Tags
Last Accessed
Revoke sessions
Permissions policies (1) Info
You can attach up to 20 managed policies.
Remove
Add permissions
Filter by Type
Search
All types
Policy name
Type
Attached entities
CloudWatchLogsFullAccess
AWS managed
1


Successfully created flow log for the following resource:
vpc-0f538babade953edd
fl-0878eecef4520af83 / MyVPC-CloudWatch-FlowLogs
Actions
Details
Flow Log ID
fl-0878eecef4520af83
Destination Type
cloud-watch-logs
Traffic Type
All
File Format
-
Name
MyVPC-CloudWatch-FlowLogs
Destination Name
/aws/vpc/flowlogs
Max Aggregation Interval
1 minute
Hive Compatible Partitions
–
State
Active
IAM Role
arn:aws:iam::676643701647:role/VPCFlowLogsCloudWatchRole
Log Format
Default
Partition Logs
–
Creation Time
Thursday, October 29, 2026 at 17:19:55 GMT+5:30
Cross Account IAM Role
–
Tag field specifications
-
Tags
Integrations
Tags
Manage tags
Search tags
Key
Value


Successfully created flow log for the following resource:
vpc-0f538babade953edd
Explore integration options
fl-064c1f91cf30e24b0 / MyVPC-S3-FlowLogs
Actions
Details
Flow Log ID
fl-064c1f91cf30e24b0
Destination Type
s3
Traffic Type
All
File Format
Plain text
Name
MyVPC-S3-FlowLogs
Destination Name
my-vpc-flow-logs-ayesha-2026
Max Aggregation Interval
1 minute
Hive Compatible Partitions
Not enabled
State
Active
IAM Role
–
Log Format
Default
Partition Logs
Daily
Creation Time
Thursday, October 29, 2026 at 17:22:41 GMT+5:30
Cross Account IAM Role
–
Tag field specifications
-
Tags
Integrations
Tags
Manage tags
Search tags
Key
Value


VPCs
VPC encryption controls
Your VPCs (1/2) Info
Last updated 5 minutes ago
Actions
Create VPC
Find VPCs by attribute or tag
Name
VPC ID
State
Encryption c...
Encryption control ...
Block Public...
IPv
My-vpc
vpc-0f538babade953edd
Available
–
–
Off
10.
–
vpc-0a49cd508e18bd177
Available
–
–
Off
172
vpc-0f538babade953edd / My-vpc
Details
Resource map
CIDRs
Flow logs
Tags
Related resources
Integrations
Flow logs (2)
Actions
Create flow log
Search
Name
Flow log ID
Traffic type
Destination type
MyVPC-CloudWatch-FlowLogs
fl-0878eecef4520af83
All
cloud-watch-logs
MyVPC-S3-FlowLogs
fl-064c1f91cf30e24b0
All
s3



1. I selected my **VPC (`My-vpc`)** in the AWS VPC Console.
2. I created an **S3 bucket** to store the VPC Flow Logs.
3. I created a **CloudWatch Log Group** to store and monitor the flow logs.
4. I created an **IAM role** with the required permissions for VPC Flow Logs to send logs to CloudWatch.
5. I enabled **VPC Flow Logs** for my VPC.
6. I selected **All traffic** to capture both accepted and rejected network traffic.
7. I configured one flow log to send the logs to the **S3 bucket**.
8. I configured another flow log to send the logs to the **CloudWatch Log Group**.
9. I verified that the flow logs were successfully created and were in the **Active** state.
10. I generated some network traffic and verified that the logs were appearing in **S3 and CloudWatch**.